

Model vs Index — CAGR Comparison

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What period this covers

Item	Value
Test window	2018-11-01 to 2020-06-30
Rebalance frequency	monthly
Rebalance dates	19
Elapsed time	1.58 years
Holding period	entry to the next rebalance's entry (chaining)
Portfolio	equal-weighted top 20, rebalanced each period
CAGR formula	product of (1 + period return) over 19 periods, then raised to 1/1.58

Basis

Price return compared with price return. NSE's historical index endpoint serves price indices only — every total-return (TRI) spelling returns no data. The strategy's dividends are therefore reported separately rather than folded into the headline, because comparing a dividend-inclusive strategy against a dividend-less index would credit the model with roughly the market's yield (about 1–1.5% a year in India) for nothing.

Index returns are measured over the sessions the strategy actually traded — the same next-session entry and the same exit — not between month-ends, so neither side is measuring a different span of market time.

Headline: CAGR by strategy

Strategy	Gross	Net	vs UNIVERSE	vs N50	vs N500	vs NMIDCAP 150	vs NSMALLCAP 250
equal weight universe	-10.95%	-12.81%	-1.86%	-10.16%	-8.80%	-6.72%	+0.73%
growth only	-16.02%	-19.94%	-8.98%	-17.28%	-15.93%	-13.85%	-6.40%
model 5	19.39%	11.05%	+22.00%	+13.70%	+15.06%	+17.14%	+24.59%
model 5 gated	19.39%	11.05%	+22.00%	+13.70%	+15.06%	+17.14%	+24.59%
momentum only	10.71%	1.78%	+12.73%	+4.43%	+5.79%	+7.87%	+15.32%
quality only	-5.31%	-11.10%	-0.15%	-8.45%	-7.09%	-5.01%	+2.44%
random ranking	-1.10%	-23.82%	-12.87%	-21.17%	-19.81%	-17.73%	-10.28%
simple quality momentum	7.89%	0.44%	+11.39%	+3.09%	+4.45%	+6.53%	+13.98%
value only	21.24%	16.45%	+27.40%	+19.10%	+20.46%	+22.54%	+29.99%

vs UNIVERSE is the column that matters, and it is deliberately placed before the index columns. It compares the strategy against an equal-weighted holding of the very same point-in-time universe it selected from, so it isolates stock selection.

Why the index columns flatter every strategy. The eligible universe returned -10.95% against NIFTY 500's -4.01% — a -6.95 point gap earned purely by equal-weighting a broad, small-cap-heavy universe, before any selection at all. That gap is inside every "vs index" number below. A strategy that beats NIFTY 500 by 20 points while matching the universe has selected nothing.

Index reference

Index	CAGR	Volatility	Sharpe	Max drawdown	Periods
NIFTY 50	-2.65%	27.84%	0.05	-32.25%	19
NIFTY 500	-4.01%	28.16%	0.01	-31.85%	19
NIFTY MIDCAP 150	-6.09%	31.34%	-0.03	-32.75%	19

NIFTY SMALLCAP 250	-13.54%	35.09%	-0.22	-42.26%	19
Eligible universe (equal weight)	-10.95%	-	-	-	19

The **eligible universe** row is the benchmark that controls for size and breadth: the same point-in-time universe the strategy chose from, held equally weighted. Beating NIFTY 500 while trailing this means the model captured a small-cap tilt rather than stock selection.

Risk-adjusted comparison

Strategy	Basis	CAGR	Volatility	Sharpe	Max DD	Hit rate
equal weight universe	net	-12.81%	36.05%	-0.19	-43.18%	58%
growth only	net	-19.94%	43.40%	-0.29	-54.60%	47%
model 5	net	11.05%	26.70%	0.52	-17.53%	58%
model 5 gated	net	11.05%	26.70%	0.52	-17.53%	58%
momentum only	net	1.78%	25.01%	0.19	-21.27%	53%
quality only	net	-11.10%	38.45%	-0.11	-42.63%	47%
random ranking	net	-23.82%	38.23%	-0.50	-52.10%	32%
simple quality momentum	net	0.44%	24.50%	0.14	-23.92%	42%
value only	net	16.45%	47.18%	0.54	-33.56%	47%

Excess return against NIFTY 500

Strategy	Mean excess/period	Tracking error	Information ratio	Periods beaten
equal weight universe	-0.60%	13.95%	-0.51	26%
growth only	-1.06%	21.97%	-0.58	32%
model 5	+1.15%	13.72%	1.01	53%
model 5 gated	+1.15%	13.72%	1.01	53%
momentum only	+0.39%	17.51%	0.26	37%
quality only	-0.36%	17.90%	-0.24	37%
random ranking	-1.61%	17.11%	-1.13	42%
simple quality momentum	+0.26%	17.63%	0.18	58%
value only	+2.11%	30.37%	0.84	58%

Information ratio is excess return per unit of tracking error — how reliably the outperformance was earned, not how large it was. **Periods beaten** is the share of rebalances where the strategy outpaced the index; a high CAGR with a low share was carried by a handful of periods.

Turnover and cost drag

Strategy	One-way turnover	Gross CAGR	Net CAGR	Cost drag
equal weight universe	0.055	-10.95%	-12.81%	1.86%
growth only	0.158	-16.02%	-19.94%	3.92%
model 5	0.362	19.39%	11.05%	8.34%
model 5 gated	0.362	19.39%	11.05%	8.34%
momentum only	0.468	10.71%	1.78%	8.93%
quality only	0.300	-5.31%	-11.10%	5.79%
random ranking	0.958	-1.10%	-23.82%	22.72%
simple quality momentum	0.398	7.89%	0.44%	7.45%
value only	0.158	21.24%	16.45%	4.79%

Cost drag is annualised, and it is the difference between a ranking that works on paper and one that works in an account. A strategy replacing most of its book every month pays it repeatedly.

Before quoting these numbers

An index is investable and this portfolio is a simulation. The index CAGR is achievable through a low-cost fund; the strategy CAGR assumes every fill happened at the modelled price and size.

The window is what it is. A CAGR over a few years is dominated by the market regime inside it. Check the periods-beaten share and the drawdown before treating a CAGR gap as skill.

Costs are modelled, not measured. Statutory charges are exact and effective-dated; half-spread and market impact are assumptions, which is why gross and net are both shown.

Delisting recovery is an assumption. Re-run with --delisting-strategy zero and last_close to see how much of the gap depends on it.